

Neoadjuvant Chemoimmunotherapy in Resectable Non-small Cell Lung Cancer

Yulong Chen^{1#}, Bo Yan^{2#}, Zhenzhen Hui^{3#}, Gang Zhao⁴, Jie Liu⁵, Huan Zhang⁶, Ziqing Zeng³, Ran Zhang⁷,
Xiubao Ren³, Jian You¹

¹Department of Lung Cancer, Tianjin Lung Cancer Center, Tianjin Medical University Cancer Institute and Hospital, National Clinical Research Center for Cancer, Key Laboratory of Cancer Prevention and Therapy, Tianjin's Clinical Research Center for Cancer, Tianjin 300060, China; ²Department of Radiotherapy, National Clinical Research Center of Cancer, Tianjin Medical University Cancer Institute and Hospital, Tianjin 300060, China; ³Immunology Laboratory, National Clinical Research Center of Cancer, ⁴Department of Pathology, National Clinical Research Center of Cancer, Tianjin Medical University Cancer Institute and Hospital, Tianjin 300060, China; ⁵College of Computer and Control Engineering, College of Software, Nankai University, China; ⁶Cancer Prevention Center, Tianjin Medical University Cancer Institute and Hospital, National Clinical Research Center for Cancer, Key Laboratory of Cancer Prevention and Therapy, Tianjin, Tianjin's Clinical Research Center for Cancer, Tianjin 300060, China; ⁷Department of Minimally Invasive Surgery, Taishan Cancer Hospital, Tianjin 300308, China.

[#]These authors contributed equally to this work.

Correspondence to: Jian You. Department of Lung Cancer, Tianjin Lung Cancer Center, Tianjin Medical University Cancer Institute and Hospital, Huan-Hu-Xi Road, Ti-Yuan-Bei, He Xi District, Tianjin 300060, China. Email: youjiancn@126.com; Xiubao Ren. Immunology Lab, National Clinical Research Center of Cancer, Tianjin Medical University Cancer Institute and Hospital, Tianjin 300060, China. E-mail: renxiubao@tjmuch.com.

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ABSTRACT

Background: A subset of patients experience objective clinical benefit after neoadjuvant PD-1 blockade in resectable lung cancer. The combining checkpoint blockade and chemotherapy as neoadjuvant chemoimmunotherapy therapeutic regimen might be more effective but has not been tested in resectable non-small cell lung cancer (NSCLC).

Methods: This is an investigator-initiated trial for patients with resectable IIIA and IIIB non-NSCLC. The primary objective was pathological complete response (pCR). Also, the overall survival (OS) and progression-free survival (PFS) were analyzed. The correlation between the pathology response and some clinical factors was studied to identify some predictors, including PD-L1 expression. Also, artificial intelligence was utilized to incorporate multiple factors and develop a high-performance prediction model. Multiple machine learning models were built and the roles of different indicators were assessed, while the predictability of treatment outcomes was validated.

Findings: Neoadjuvant PD-1 blockade plus chemotherapy was associated with few immediate adverse events. The NCIO did not delay planned surgery and led to a complete pathological response in 56% patients and a major pathological response in 80% patients. No significant correlation between the pathological response and PD-L1 expression. However, significant difference was observed between the Invade and no Invade status, with 83.3% pCR and 100% MPR rate vs. 38.5% pCR and 61.5% MPR rate ($P=0.007$). After model training and optimization, as well as 5-fold cross-validation, SVM and KNN were confirmed with higher prediction accuracy (both 88%).

Interpretation: Neoadjuvant PD-1 blockade plus chemotherapy was associated with few side effects, and induced a complete pathological response in 56% patients. The status of the trachea was associated with pathological response. The machine learning method captures the internal correlation between the detection index and pathological response to construct a predictable model.

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Keywords: neoadjuvant chemoimmunotherapy; non-small cell lung cancer; PD-1; artificial intelligence; surgery

RESEARCH IN CONTEXT

Evidence before this study

PubMed was searched using the search strings (no search restrictions): “PD-1 OR PD-L1 OR pembrolizumab OR MK- 3475 OR lambrolizumab OR nivolumab OR BMS-936558 OR MPDL3280A OR atezolizumab OR BMS-936559 OR MEDI4736 OR durvalumab OR avelumab AND lung cancer and Neoadjuvant Chemotherapy” or “PD-1 OR PD-L1 OR pembrolizumab OR MK- 3475 OR lambrolizumab OR nivolumab OR BMS-936558 OR MPDL3280A OR atezolizumab OR BMS-936559 OR MEDI4736 OR durvalumab OR avelumab AND lung cancer and Neoadjuvant chemoimmunotherapy”. One case report described the results of a patient with perioperative chemoimmunotherapy. We also identified a brief report, which studied the feasibility of sleeve lobectomy after neoadjuvant chemoimmunotherapy in non-small cell lung cancer. The use of a PD-1 inhibitor combined with chemotherapy as neoadjuvant chemoimmunotherapy in resectable non-small cell lung cancer and the artificial intelligence to predict the pathological response has not yet been

reported.

Added value of this study

To the best of our knowledge, this is the first report of the activity and safety of an anti-PD-1 antibody combined with and chemotherapy used as neoadjuvant treatment in patients with resectable non-small cell lung cancer. We included patients with IIIa and IIIb tumors that were difficult to be resected thoroughly. The pCR occurs irrespective of the tumor size, TNM stage, age, PD-L1 expression, and pathology type. In addition, artificial intelligence, SVM, and KNN model predict the pathological response with prediction accuracy 88%.

Implications of all the available evidence

Patients with stage IIIA to IIIB NSCLC account for a large proportion of lung cancer, and most of them exhibited high PS score and optimal immune state but poor survival due to limited treatment method. The neoadjuvant chemoimmunotherapy achieves a high pCR, which might prolong the survival, and even cure the cancer. The clinical benefit and acceptable safety profile observed in this study suggested that neoadjuvant PD-1 blockade plus chemotherapy is an effective treatment option for patients who cannot benefit from standard therapy.

INTRODUCTION

The majority of the treatment methods have focused on the establishment of new treatment options for non-small cell lung cancer (NSCLC). Programmed cell death 1 (PD-1) and programmed cell death ligand 1 (PD-L1) immune checkpoint inhibitors (ICIs) are currently the most investigated immunotherapies [1]. Therapeutically blocking this inhibitory molecular axis using specific monoclonal antibodies targeted to PD-1 and PD-L1 activates the immune system to recognize and target the cancer cells via a T cell-mediated immune response [2-3]. Recently, the neoadjuvant PD-1 blockade has also been investigated in resectable lung cancer but was not found to be impressive. Clinical trials LCMC3 (ClinicalTrials.gov numbers: NCT02927301) has reported that only 21% of the patients present major pathological response (MPR), while 5% exhibited pathological complete response (pCR) of the resected tumors [4]. Thus, the combination of checkpoint blockade and chemotherapy explores the potential for synergistic immune activation. Herein, we examined the safety and feasibility of the use of neoadjuvant PD-1 antibody plus chemotherapy in a small group of patients with resectable NSCLC. In addition, the correlation between the pathology response and some clinical factors was established to find some predictors, including PDL1 expression, CD4+ and CD8+ T cells and Treg cell, and invasion status of the trachea. We also used artificial intelligence to analyze the incorporation of multiple factors and develop a high-performance prediction model. If patients with a low-risk of tumor residue are identified, additional surgery after NCIO may not be an acceptable option.

MATERIAL AND METHODS

Patients

This is an investigator-initiated trial for patients with resectable IIIA and IIIB NSCLC at Tianjin Medical University Cancer Institute and Hospital (TJMUCH), Tianjin, China. The clinical stage was predicted according to the American Joint Committee on Cancer (AJCC) 8th edition. Patients received 2 courses of Pembrolizumab 100 mg IV q3w added to cisplatin 75 mg/m² IV q3w plus Pemetrexed 500 mg/m² IV q3w or paclitaxel liposome 135 mg/m² q3w. The surgery was performed 4 weeks after the last dose.

Our primary endpoint would be reached if 12/25 (48%) planned patients have a pCR. The study was focused on the effect, safety, and feasibility of the therapy. The key secondary and exploratory research content was radiological and pathological responses to treatment and immunological and pathological correlations of the response in the tumor. All the patients were monitored for adverse events, according to the National Cancer Institute Common Terminology Criteria for Adverse Events, version 4.0 [5]. All the patients underwent baseline tumor staging, including pretreatment biopsy, pathological evaluation of mediastinal lymph nodes (if indicated) by bronchoscopy or mediastinoscopy, positron-emission tomography-computed tomography (PET-CT), and contrast-enhanced CT or magnetic resonance imaging of the brain and chest; chest CT was repeated within 7 days before the surgery. Changes in the tumor size were evaluated according to the Response Evaluation Criteria in Solid Tumors (RECIST) version 1.1 [6]. The resection of the primary tumor and lymph nodes was carried out according to the institutional standards. Feasibility was defined as any delay in the planned surgery of <30 days. This study was approved by the institutional review boards at the Tianjin Medical University Cancer Institute and Hospital.

Immunohistochemistry (IHC)

PD-L1 was analyzed by IHC at the Pathology Department of TJMUCH using the Monoclonal Mouse Anti-human PD-L1 clone 22C3 (LOT10145059, Dako, Carpinteria, CA, USA). Tumor proportion score (TPS) and combined positivity score (CPS) were calculated as the percentage of at least 100 viable cells with complete or partial membrane staining. However, in patients with no residual tumor after the treatment, only CPS was calculated. The interpretation of TPS or CPS was provided by the pathologist of TJMUCH.

Flow cytometry

CD45, CD3, CD4, CD8, CD127, and CD25 were purchased from BioLegend (San Diego, CA, USA). The antibodies were as follows: FITC anti-human CD4 317408; PE anti-human CD25 302606; PE anti-human CD163 326506; PerCP/Cy5.5 anti-human CD127 (IL-7R α) 351322; APC anti-human CD68 333810; Brilliant Violet 421 anti-human CD3 300434; Brilliant Violet 510 anti-human CD45 304036. The cells were stained with optimal amounts of each antibody for 30 min at 4 °C in the staining buffer, according to the manufacturer's protocol. Data (10,000 events) were acquired on BD-Aria II flow cytometer (BD Biosciences, San Diego, CA, USA) and analyzed using FlowJo software (FlowJo LLC, Ashland, OR, USA).

Pathological assessments

Primary lung tumor and lymph node surgical specimens were staged according to the criteria of the AJCC(7th edition) for evaluating the tumor size, affected lymph nodes, and metastases. Primary tumors were assessed

for the percentage of residual viable tumor that was identified on routine hematoxylin and eosin (HE) staining, and tumors with <10% viable tumor cells were considered to have a major pathological response (MPR), and no viable tumor cells were deemed as pCR [1,7-8].

Artificial intelligence

Based on the data from these 25 cases, we performed a statistical analysis of the importance of indicators and modeled the pathological response prediction as a binary classification problem. Next, we applied representative models, such as Gaussian Naive Bayes (Gaussian NB), K nearest neighbor classifier (KNN), linear discriminant analysis (LDA), support vector machine (SVM), decision tree classifier, and logistic regression. A variety of machine learning algorithms were used for training and optimizing the models. The 5-fold cross-validation was applied to evaluate the predicted results and assess the statistical significance with the limited data.

Statistical analysis

Mann–Whitney test was used to compare the pathological response between Invasion and no Invasion status and TPS score. Independent sample t-test was used to compare the proportion of CD4, CD8, and Treg cells between genotype cohorts. PFS and OS was estimated using Kaplan–Meier curve and log-rank test. Pearson’s and Spearman’s correlation coefficient were used to analyze the correlation between TPS score or TIST and pathological response, respectively. All P-values were based on a two-sided hypothesis, and data were analyzed using SPSS 22.0.

Role of the funding source

This study did not receive any specific grant from any funding agency in the public, commercial, or not-for-profit sectors. Dr. Jian You designed the study. All authors contributed to the interpretation of the data. Dr. Chen prepared the manuscript. All authors had full access to the data and the responsibility for the decision to submit the article for publication.

RESULTS

Safety and feasibility

From January 2019 to May 2020, 25 patients were enrolled, and all had undergone surgery. The median age of the cohort was 62.2 ± 6.4 (43–72) years, 84% were men, 80% had squamous carcinoma, 84% were stage IIIA, and 56% were PD-L1 positive. Our primary endpoint was achieved by 12/25 (48%), 14/25 (56%) achieved pCR, and 20/25 (80%) exhibited MPR. In the pCR group, the median age was 61.07 ± 7.6 years, 92.8% were men, 92.8% had squamous carcinoma, 92.8% were stage IIIA, and 64.29% were PD-L1 positive, while in the non-pCR group, the median age was 63.64 ± 4.39 years, 72.72% were men, 63.63% had squamous carcinoma, 72.72% were stage IIIA, and 45.45% were PD-L1 positive (Table 1).

Neoadjuvant Pembrolizumab plus paclitaxel liposome or pemetrexed combined with cisplatin was not associated with any previously unreported toxic effects. Pre-surgical grade 3 toxicity, i.e. rash, occurred in 1/25 patients. The median interval between the administration of the second dose of NCIO and surgery was

33.4 (range, 28–35) days, and all the 25 eligible patients (100%) underwent complete tumor resection. No treatment-related surgical delays occurred as defined in the protocol.

Representative radiological and pathological responses after two preoperative NCIO are shown in Figure 3. According to the RECIST criteria, 13 (52%) patients had a partial response, and 12 (48%) had stable disease. 19/25 (76%) patients, who underwent resection, pathological downstaging from the pretreatment clinical stage was observed (Table 1, Figure 1).

At a median of 10.2 (range, 3–19) months of postoperative follow-up, 23/25 (92%) patients who had undergone surgical resection were alive and recurrence-free. One patient died 3 months after the surgery because of cerebral ischemic stroke (patient 3), and the other died after 10 months post-surgery due to mass N2 lymph nodes metastasis (patient 1). One patient was diagnosed with brain metastasis (patient 15). The median duration of recurrence-free survival had not been reached at the time of data analysis. Hence, based on the available data, no significant differences were detected in the progression-free survival (PFS) and overall survival (OS) between the pCR and non-pCR groups (Figure 2).

Pathological findings after neoadjuvant PD-1 blockade

MPR occurred in 20/25 (80%) patients; 14 patients (56%) had pCR in the primary tumor. However, 1/14 patient had a residual tumor in hilar lymph nodes (patient 15). The median degree of pathological regression in the primary tumor was -81.88% (range, -100 to 0). Despite apparent tumor enlargement on preoperative CT (possibly because of the infiltration of immune cells into the tumor), one patient exhibited pCR (patient 21). In primary tumors with MPR, we observed a large number of infiltrating lymphocytes and macrophages. This finding was compatible with an immunological mechanism of response along with the phenomenon that necrotic tumor was associated with fibrotic tissue repair (Figure 3).

Invasion status of the trachea with pathological response

We also analyzed the TIST by bronchoscope based on CT or PET-CT for all the patients. If the tumor invades the trachea, a neoplasm is seen in the trachea, and bleeding occurs easily by the touch of the bronchoscope. If the tumor has not invaded the trachea, we can only observe the red color and narrowing near the opening of the trachea due to the pressure from the tumor (Figure 4a). Interestingly, 12/25 (48.0%) patients were evaluated as Invade and 13/25 (52.0%) as no-Invade. Of these, 10 (83.3%) patients in the Invade group presented pCR response, and all the 12 (100%) patients showed the MPR response. On the other hand, only 5 (38.5%) patients in the no-Invade group showed pCR response, while 8 (61.5%) patients presented MPR, and 5 (38.5%) patients did not demonstrate MPR ($P=0.007$). Next, we used Spearman's coefficient of rank to analyze the correlation between the invasion status and the pathological response (Spearman's $r=0.551$, $P=0.004$; Figure 4b).

Expression of PD-L1 with pathological response

The expression of PD-L1 could be evaluated in pretreatment biopsy samples in all patients (Figure 5A). A pCR or MPR response occurred in both PD-L1-positive and PD-L1-negative tumors. The correlation between

the TPS scores of pretreatment biopsy samples and pathological regression was analyzed. Figures 5B shows that there is no association between PD-L1 expression before the treatment and the pathological response to NCIO (Pearson's $r=-0.188$; $P=0.368$).

The expression of PD-L1 was also evaluated in post-surgery samples in all patients. The TPS changes before and after the treatment were also evaluated. As shown in Figure 5C, 9/14 patients showed elevated TPS, and 5 showed a decrease in the score in the pCR group (TPS was replaced by CPS due to no tumor residual). The portion of the TPS increase was 64.28%, and that of the decrease was 27.27% in the pCR group. In the non-pCR group, 2/11 patients exhibited a TPS increase, and 3 patients showed a decrease. Moreover, 6/11 patients showed <5% changes in the TPS. Thus, the proportion of the TPS increase was 18.18%, and that of the decrease was 27.27% in the non-pCR groups ($P<0.01$).

Artificial intelligence

In order to extract the features related to the target task and exclude the noise influence of irrelevant feature fields, we first used the F test to analyze whether each feature had significant changes under different values of the target field, and then completed the correlation assessment of each field. The visualized importance of features on different tasks is shown in Figure 6A.

Based on the important analysis of the indicators, a variety of machine learning models were used to construct the pathological response prediction models. Following 100% pathological regression as the prediction target and index information as the input characteristic, we modeled this problem as a binary classification problem. Several machine learning algorithms, such as The Gaussian NB, KNN, LDA, SVM, Decision Tree Classifier, and logistic regression, were used for training and optimizing the models. Also, the 5-fold cross-validation was applied to evaluate the predicted results. As shown in Figure 6B, the modeling SVM and KNN were confirmed to utilize the comprehensive information of multiple indicators, and the prediction accuracy was 88%, which was better than other modeling tools.

Immune proofing of T cells to NCIO

To further explore the T-cell proofing of these patients, flow cytometry analysis was performed. CD4⁺ and CD8⁺ T cells and Treg cells were evaluated in the post-surgery samples in 17 available patients (Figure 7). The ratio of the CD4⁺, CD8⁺ T cells, and Treg cells to T cells and all cells of the sample were analyzed in the pCR and MPR groups. As shown in Figure 7b, the proportion of CD 4⁺ T cells to T cells was 36.93 ± 12.34 and 43.67 ± 14.61 in the pCR group and non-PCR group, respectively ($P>0.05$), while the percentage of CD 4⁺ T cells to all cells in the sample was 7.42 ± 4.94 and 10.39 ± 11.90 in the pCR and non-PCR groups, respectively ($P>0.05$). Moreover, the percentage of CD8⁺ T cells to T cells was 50.74 ± 12.78 and 43.90 ± 19.70 in the pCR and non-PCR groups, respectively ($P>0.05$). Furthermore, the percentage of CD 8⁺ T cells to all cells in the sample was respectively 10.43 ± 7.97 and 9.71 ± 6.97 in the pCR and non-PCR groups, respectively ($P>0.05$). The percentage of Treg cells to T cells was 1.55 ± 1.25 and 3.57 ± 2.69 in the pCR and non-PCR groups, respectively ($P>0.05$), while that of Treg cells to all cells in the sample was 0.41 ± 0.57 and 1.04 ± 1.27 in the pCR and non-PCR groups, respectively ($P>0.05$).

DISCUSSION

We observed that neoadjuvant administration of two doses of Pembrolizumab plus paclitaxel liposome or pemetrexed combined with cisplatin in patients with IIIA-IIIB resectable NSCLC was associated with few immediate adverse events. NCIO did not delay the planned surgery and led to pCR in 56% patients and MPR in 80% patients. Also, it was not associated with any previously unreported toxic effects [1-4,9]. The NADIM study reported that 13% of the patients encountered G3–5 TRAEs, and the most common postsurgical complication was respiratory infections [9]. In the current study, only one patient encountered G3–5 TRAE, which is rash, and none of the patients experienced postsurgical complications, including respiratory infections. Since it is not cost-effective, a 100 mg i.v. dose of pembrolizumab was administered over 30 min every 3 weeks, which was lower than the recommended dose. The current results demonstrated that this lower dose was effective in treatment-related fewer adverse events.

Immune-checkpoint blockade derived long-term OS in a subset of cancer patients [10-11]. Strikingly, only a subset of patients with advanced solid tumors responded to the treatment. Therefore, developing a method to identify patients who are most likely to respond to immunotherapy is essential. Several predictors, such as PD-L1 [12-14], tumor mutational burden (TMB) [15], and radiomics [16], were explored to help in the prediction of clinical outcomes. However, there is yet no reliable predictor to aid in clinical decision-making. Based on our findings, no significant correlation could be established between the PD-L1 expression and pathological regression. Also, the pathological regression did not differ significantly between the TPS scores $\geq 50\%$ and $< 50\%$ (Spearman's $r=0.218$ and $P=0.294$). The pCR or MPR occurred in both patients with high or low TPS. The PD-L1 expression is not a good predictor of the pathological response. Hence, we also studied the dynamic changes in squamous cell carcinoma (SCC), including the changes in the tumor size with respect to NCIO. These factors did not show any significant difference between the pCR and non-pCR groups. Surprisingly, TIST was detected as the pathological response. 83.3% patients in the Invade group showed pCR and 100% patients exhibited MPR, while only 38.5% patients in the no-Invade group presented pCR response, 61.5% patients showed MPR, and 38.5% patients did not exhibit MPR. Next, we used Spearman's coefficient of rank to analyze the correlation between the invasion status and the pathological response; the Spearman's r was 0.551 and $P=0.004$. These values indicated that the TIST and pathological responses were moderately correlated, which might be a valuable parameter for the prediction of the pathological response to NCIO.

Due to the extraordinary complex mechanism, individual factors cannot accurately predict the comprehensive analysis incorporating multiple factors. Besides the antigen-presenting pathway, future models predicting the clinical outcome with ICI necessitate the integration of additional factors affecting the tumor-host interaction such as PD-L1 expression, gut microbiota composition, patient germline genetics, immune microenvironment composition, as well as the absence/presence of soluble inhibitory molecules as proposed in several cancer immunograms [17-19]. We also used artificial intelligence to analyze the incorporation of multiple factors and develop a high-performance prediction model. With 100% pathological regression as the prediction target and index information as the input characteristic, we modeled this as a binary classification problem. Also, the 5-fold cross-validation was applied to evaluate the predicted results. Finally, the SVM and KNN modeling provided comprehensive information on multiple indicators with a prediction

accuracy of 88%. If patients with a low risk of the residual tumor are identified, no additional surgery after NCIO may be an acceptable option.

Furthermore, we also studied the immune proofing of T cells to the NCIO and identified CD4⁺ and CD8⁺ T cells and Treg cells. Goldberg et al. demonstrated that PD-1 blockade enhances the early stage of T-cell activation in lymph nodes (18). Liu et al. reported that neoadjuvant PD-1 blockade enhances the systemic priming of antitumor T cells, thereby potentially eliminating micrometastatic cancer that might otherwise cause postsurgical relapse (19). Our findings did not reveal any significant differences in CD4⁺ and CD8⁺ T cells and Treg cells between the pCR and non-pCR groups and the MPR and non-MPR groups. The number of CD4⁺ and CD8⁺ T cells and Treg cells might be the same between the pCR and non-pCR groups. However, the function might be different as these or other immune cells caused tumor death. Salmon et al. found the importance of dendritic cells to the antitumor effects of PD-1 pathway blockade, indicating that PD-1 blockade not only worked to directly unleash the intratumoral T-cell killing but also enhanced the tumor antigen-driven priming of T cells (20). In the present study, the TPS changes before and after the treatment were also analyzed. The proportion of the TPS increase and decrease was 64.28% and 27.27%, respectively, in the pCR group, while in the non-pCR group, 2/11 patients showed an increase, and 3/11 patients showed a decrease in the score. Strikingly, in the pCR group, no tumor was left but a large number of PD-L1-positive immune cells were detected. How the anti-PD-1 drug influences the cells with PD-L1 antigen might be the key to understanding the antitumor mechanism.

Nevertheless, the present study has some limitations: the small sample size and the short postoperative follow-up period. However, this report, for the first time, demonstrated the safety of surgical resection after treatment with Pembrolizumab plus paclitaxel liposome or pemetrexed combined with cisplatin in stage IIIA to IIIB NSCLC. This study, for the first time, used artificial intelligence to analyze the incorporation of multiple factors and develop a high-performance prediction model. Thus, larger studies are essential for model training and optimization and improvement of the accuracy of artificial intelligence. Also, long-term follow-up of these studies will be necessary to decipher whether MPR and pCR rates prolong OS and PFS.

Neoadjuvant PD-1 blockade plus chemotherapy was associated with few side effects, did not delay surgery, and induced a major pathological response in 80% of the resected tumors. The current results supported the widespread use of NCIO to resectable IIIA and IIIB non-NSCLC patients. In addition, the TIST was predictive of the pathological response to PD-1 blockade, and the machine learning method captured the internal correlation between detection index and pathological response to establish a prediction model.

DATA SHARING STATEMENT

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

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CONTRIBUTORS

Dr. Jian You and Xiubao Ren derived the idea for, designed, and planned the report. Yulong Chen, Bo Yan, and Zhenzhen Hui, Ran Zhang and Ziqing Zeng acquired the data. Gang Zhao did the pathological assessments. Yulong Chen, Huang Zhang and Jie Liu analyzed the data. Jie Lu established the artificial intelligence and prediction model. All authors critically reviewed the iterations of the report and approved the final draft of the manuscript for submission.

DECLARATION OF INTERESTS

None exist.

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Table 1. Characteristics of the patients at baseline according to the pathological response

		All patients	pCR	no PCR	MPR	no MPR
Age (mean±SD, years)		62.2±6.4	61.07±7.6	63.64±4.39	61.35±6.82	65.6±2.612
Sex						
	Male	21	13	8	18	3
	Female	3	1	3	2	2
cTNM						
	IIIA	21	13	8	18	3
	IIIB	4	1	3	2	2
Smoking status						
	Never	5	2	3	3	2
	Former	2	1	1	2	0
	Current	18	11	7	15	3
Histology						
	Adenocarcinoma	3	0	3	1	2
	Squamous carcinoma	20	13	7	17	3
	Large cell carcinoma	1	0	1	1	0
	Sarcomatoid carcinoma	1	1	0	1	0
PD-L1						
	≥50%	14	9	5	12	2
	<50%	11	5	6	8	3
Surgical approach						
	Open	24	14	10	19	5
	Completed VATS§	1	0	1	1	0
Type of resection						
	Single Lobectomy #	5	3	2	3	2
	Pneumonectomy	2	1	1	1	0
	Bilobectomy\$	6	3	3	4	2
	Complex Lobectomy*	12	7	5	10	2
	Wedge resection	0	0	0	0	0

Margins of surgical resection

R0	25	14	11	20	5
R1	0	0	0	0	0
Down stage					
Yes	19	14	5	17	2
No	6	0	6	3	3

§ Video-assisted thoracoscopic surgery

Does not include pneumonectomy, bilobectomy, or complex lobectomy

\$ Does not include pneumonectomy

*Complex lobectomy includes sleeve resection and sleeve or partial resection of the pulmonary artery or surgery for pancoast tumors

Figure legends

Figure 1 Pathological assessment of response to NCIO

Pathological regression in the resected primary lung tumor after NCIO, according to the percentage of remaining viable tumor cells, for each of the 25 patients who underwent surgical resection. The gray horizontal line indicates the threshold for a major pathological response (90% regression). Clinical and pathological features that include the histology, the preoperative radiologic response (according to RECIST), gender, smoker status, and PD-L1 expression in the pretreatment biopsy sample are annotated for each patient. PR denotes partial response, SD stable disease, and PD-L1 programmed death-ligand 1.

Figure 2 Kaplan–Meier curves of PFS and OS for pCR and non-pCR groups. PFS curve (a) and OS curve (b) for all patients.

Figure 3. Patterns of radiological and pathological response to NCIO.

3A. CT imaging or PET-CT of the chest of a 56-year-old male never smoker (Patient 25) with stage IIIA squamous lung cancer before and after the administration of NCIO. In the upper panel, the pretreatment scan shows a primary tumor mass with pulmonary atelectasis in the upper lobe of the right lung. A scan performed before the surgery shows 90% shrinkage with associated tumor cavitation. The lower panel consists of the representative sections of tumor specimens obtained from this patient before the administration of NCIO and after (right) HE staining. This patient had 100% pathological regression of the large primary lung tumor and had no lymph-node metastases in the resection specimen. 3B. CT or PET-CT of the chest of a 63-year-old male smoker (Patient 5) with stage IIIA lung adenocarcinoma. In the upper panel, the tumor is stable on imaging before and after the NCIO. In the lower panel, representative sections of tumor specimens obtained from this patient before (left) and after (right) HE. Neoplastic cells are present throughout the pretreatment specimen, whereas in the posttreatment specimen, 60% tumor regression was detected.

Figure 4 Association between Invasion status of the trachea and pathological response to NCIO

Figure 4a Representative images of bronchoscopy and CT scan of Invade and No-Invade status. The Invade status means that a neoplasm could be seen in the trachea and bleeding easily by bronchoscope touch. The CT scan shows the tumor invasion into the trachea. The No-Invade status means that only the red color and narrowing near the opening of the trachea could be seen due to pressure from the tumor. The CT scan shows no tumor invasion into the trachea.

Figure 4b The association between the Invade status of the trachea and pathological response to the NCIO were analyzed. The patients with Invade status were with higher pCR and MPR rates as compared with No-Invade status, with 83.3% pCR and 100% MPR rate vs. 38.5% pCR and 61.5% MPR rate ($P=0.007$).

Figure 5. Association between PD-L1 and pathological response to NCIO

Figures 5a PD-L1 expression changes before and after NCIO. The pretreatment biopsy sample and post-surgery sample are obtained, and PD-L1 testing was performed using the Monoclonal Mouse Anti-Human PD-L1 clone 22C3. In the upper panel, the TPS score before NCIO of the patient 16 who had 100% pathological regression was 0%, and the CPS score after NCIO was 60%. In the middle panel, the TPS score

before and after NCIO of the patient 5 who had 60% pathological regression were both 100%. In the lower panel, the TPS score before NCIO of the patient 23 who had 100% pathological regression was 30%, and the CPS score after NCIO was 50%.

Figures 5B No association was observed between PD-L1 expression before the treatment and pathological response to the NCIO (Pearson's $r=-0.188$; $P=0.368$). The dashed black line indicates the linear regression line, and the dashed gray lines indicate the upper and lower boundaries of the 95% confidence interval.

Figures 4C TPS or CPS score changes before and after the NCIO in the pCR and non-pCR group. TPS increase was defined as TPS or CPS increased $\geq 5\%$; TPS decrease was defined as TPS or CPS decreased $\geq 5\%$. The proportion of the TPS increased is much higher in the pCR group as compared to the non-pCR group, 64.28% vs. 18.18%, respectively.

Figure 6a Importance of each factor used in the artificial intelligence model.

Figure 6b. (Left) The average accuracy of different models. (Right) The receiver operating characteristic (ROC) curves of the artificial intelligence models. Gaussian NB, Gaussian Naive Bayes; KNN, K nearest neighbor classifier; SVM, support vector machine; LDA, linear discriminant analysis.

Figure 7 Immune proofing of T cells in the post-surgery sample to the NCIO

Figure 7a Gating scheme for CD4⁺ and CD8⁺ T-cells and Treg cell intracellular cytokine cytometry. FSC-A, forward scatter A; SS, side scatter. Black polygons indicate gated cell subsets.

Figure 7b Proportion of CD4⁺ and CD8⁺ T-cell and Treg cells to the T cells, and all cells of the sample in the pCR group showed no significant difference as compared to the non-PCR group ($P>0.05$).

Figure 1

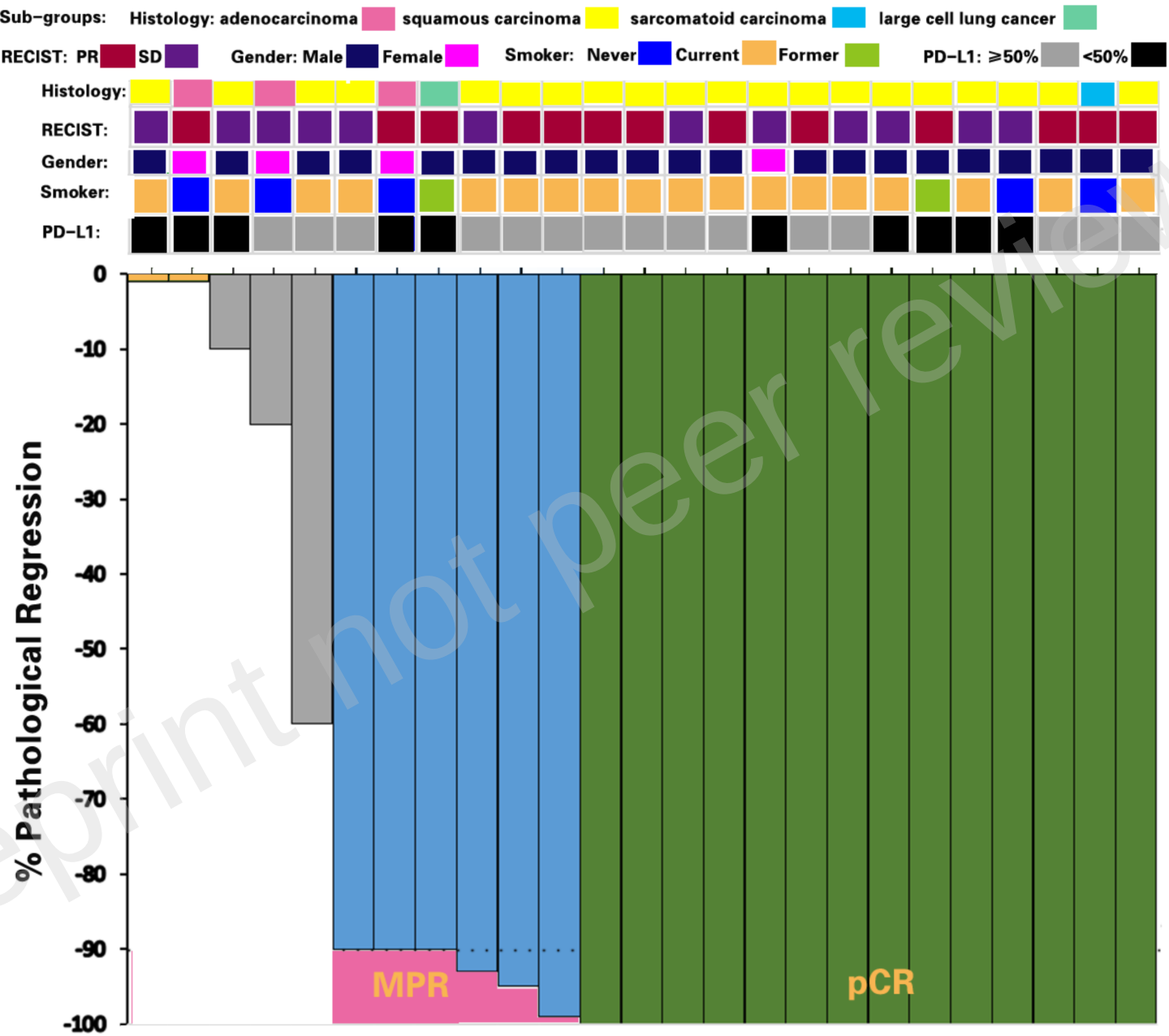


Figure 2

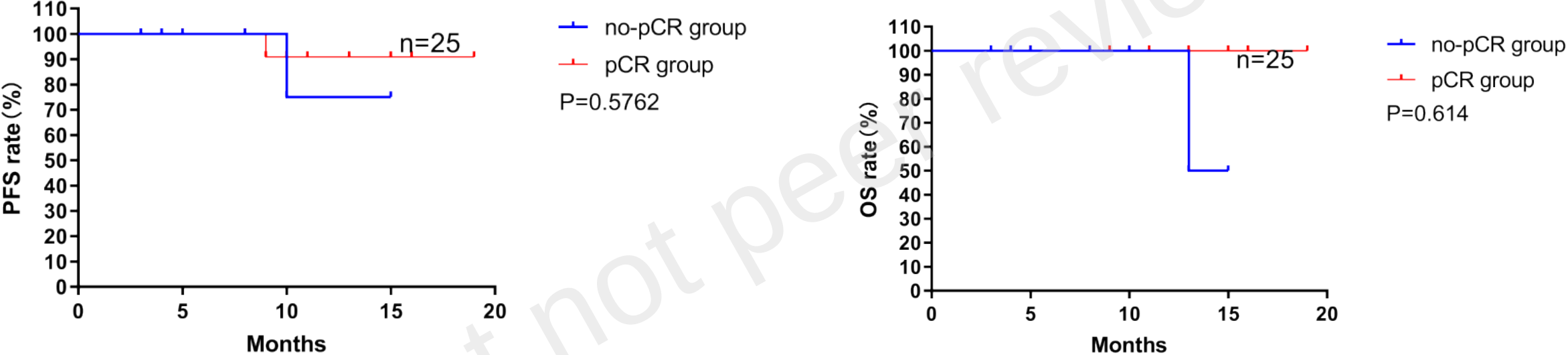


Figure 3a

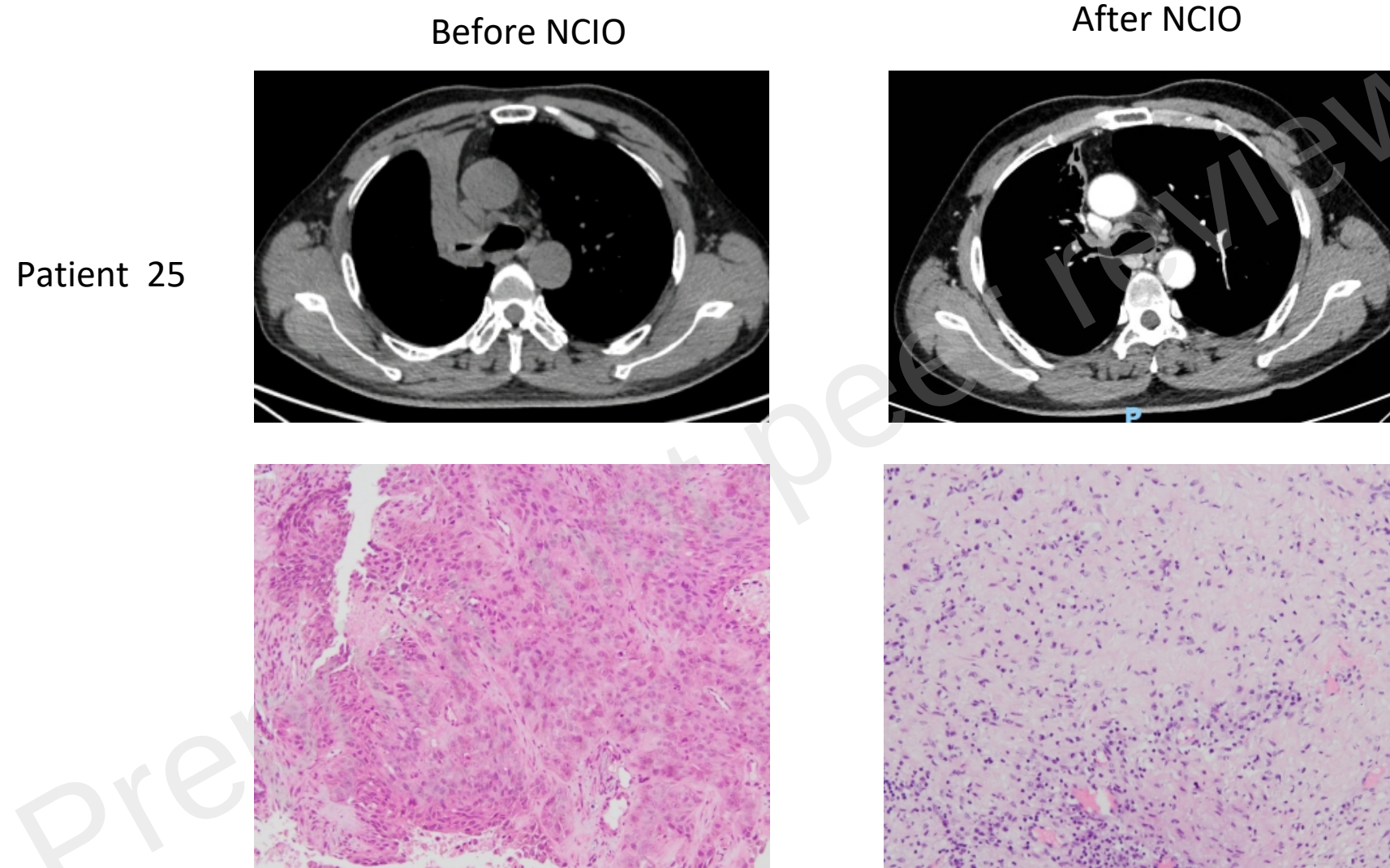
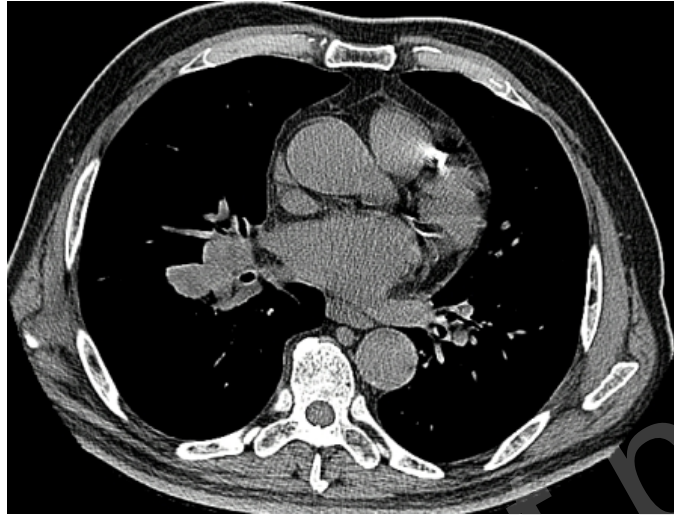


Figure 3b

Patient 5

Before NCIO



After NCIO

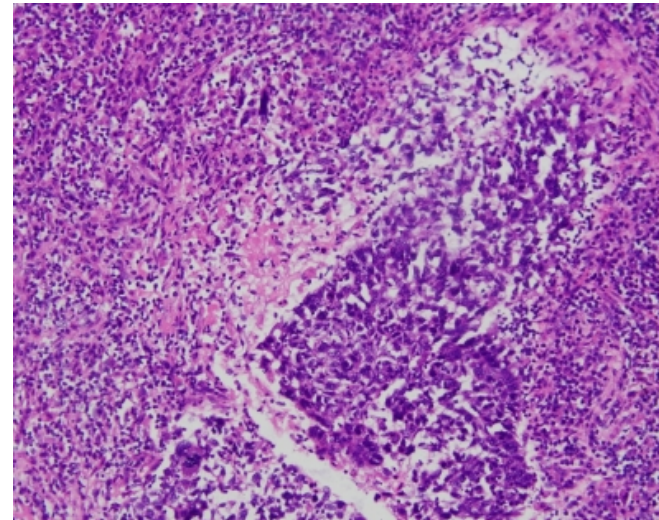
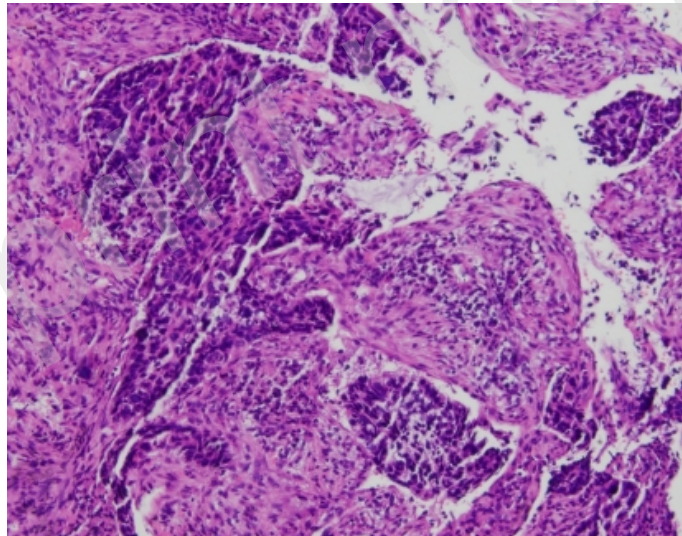
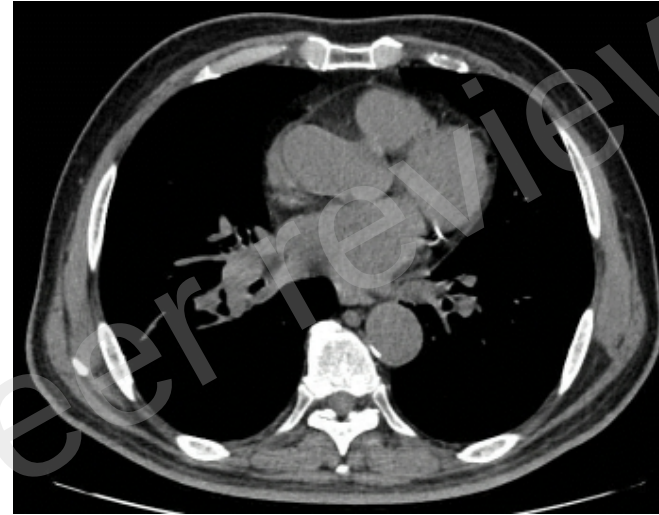
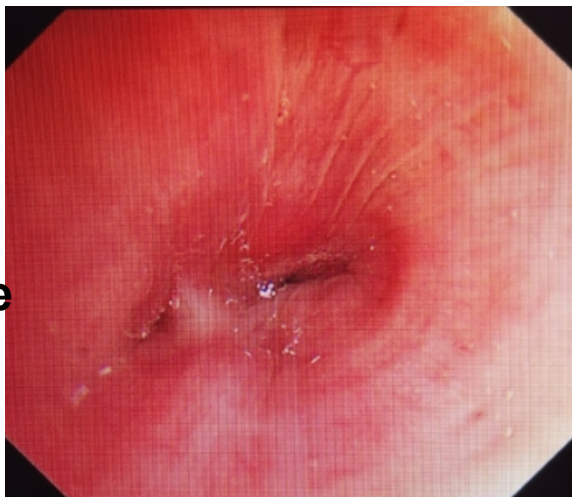


Figure 4a

No-Invade

Invade

bronchoscope



CT scan

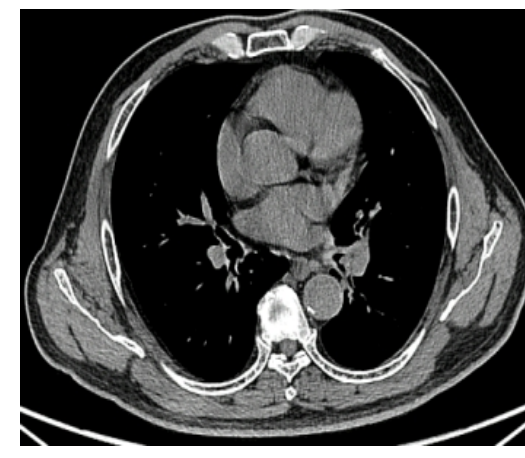
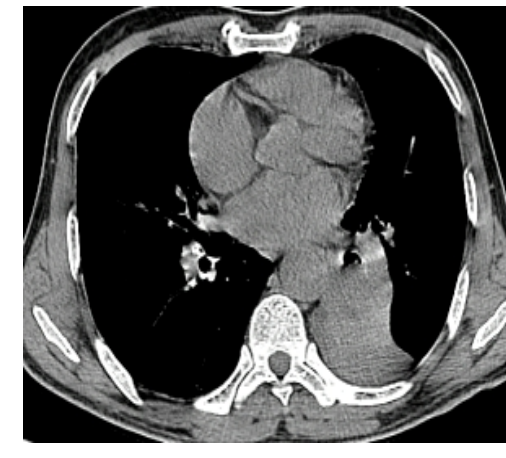


Figure 4b

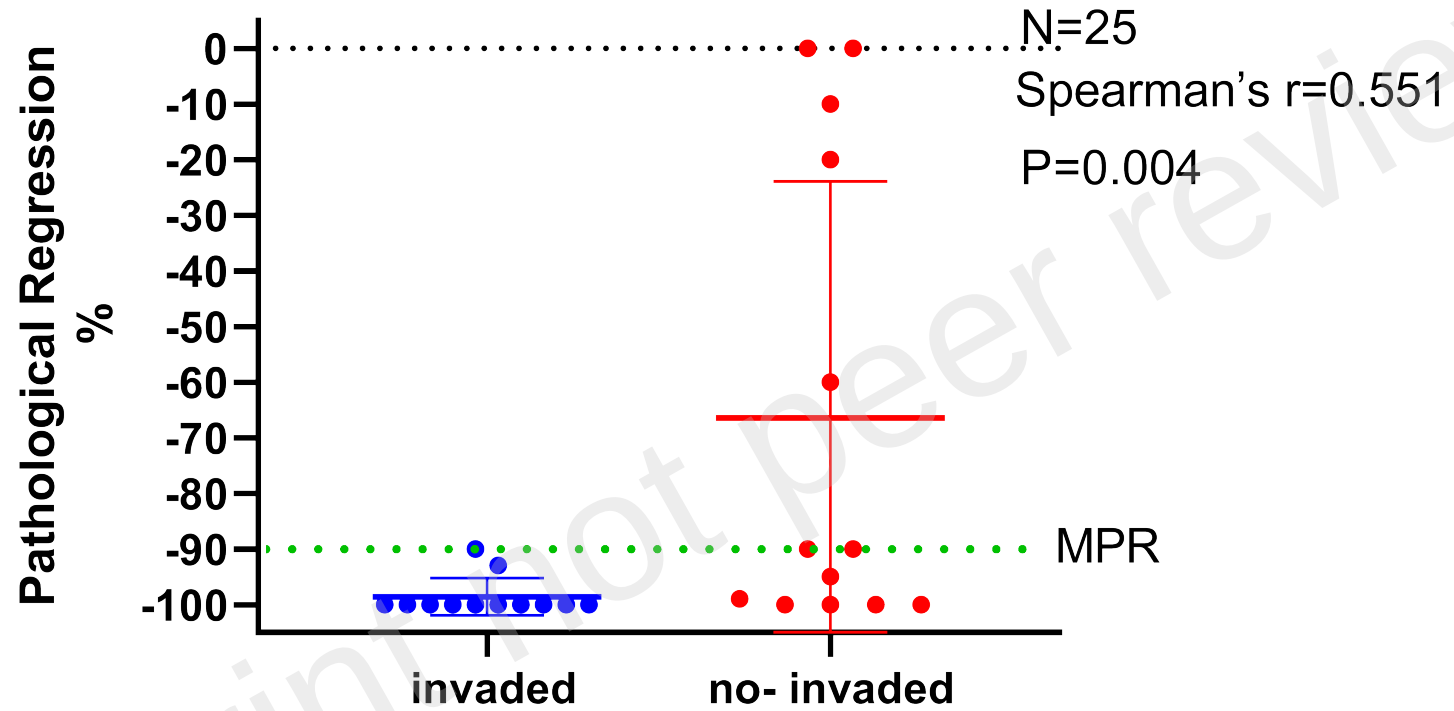
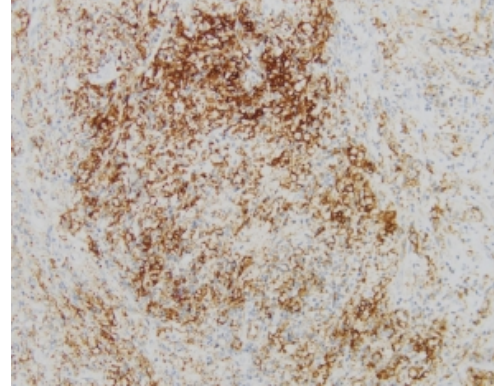
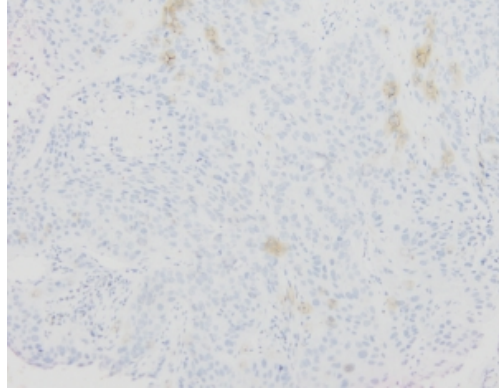


Figure 5a

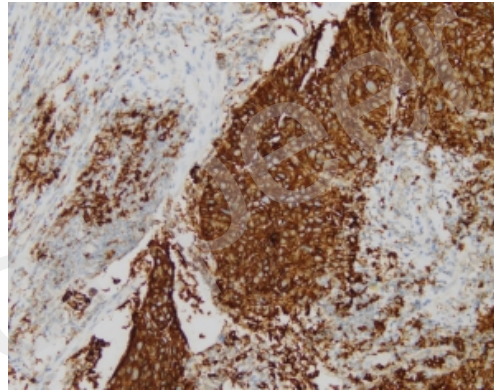
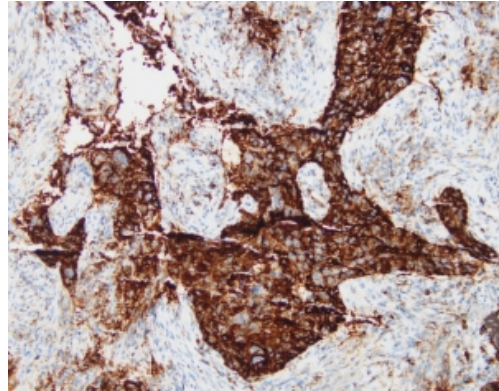
Before NCIO

After NCIO

Patient 16



Patient 5



Patient 23

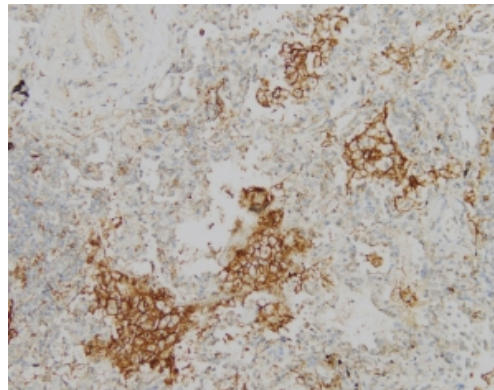
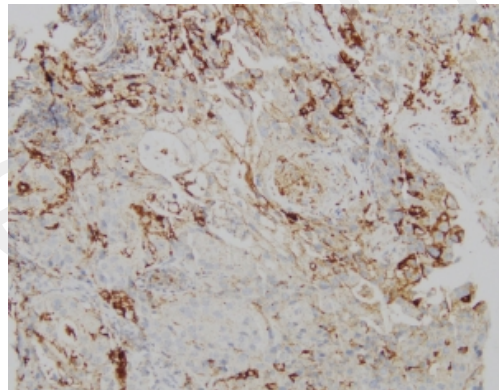


Figure 5b

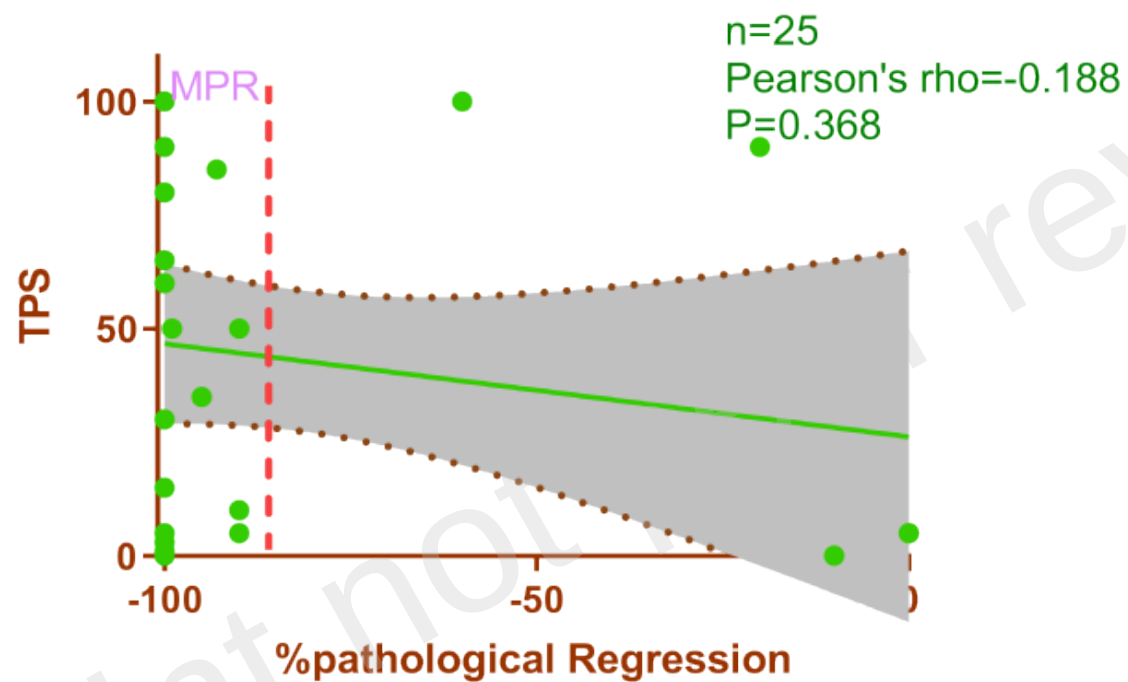


Figure 5c

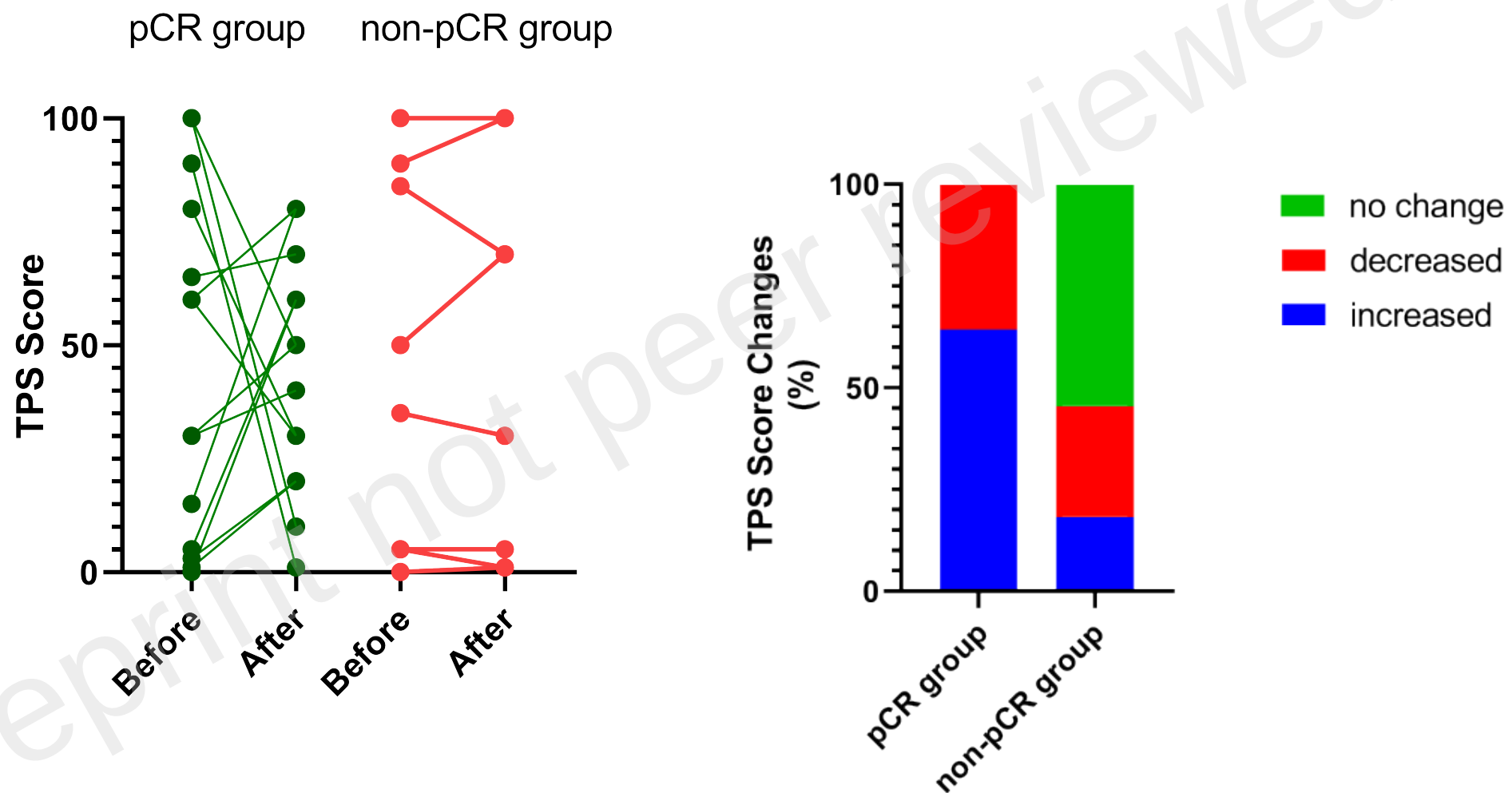


Figure 6a

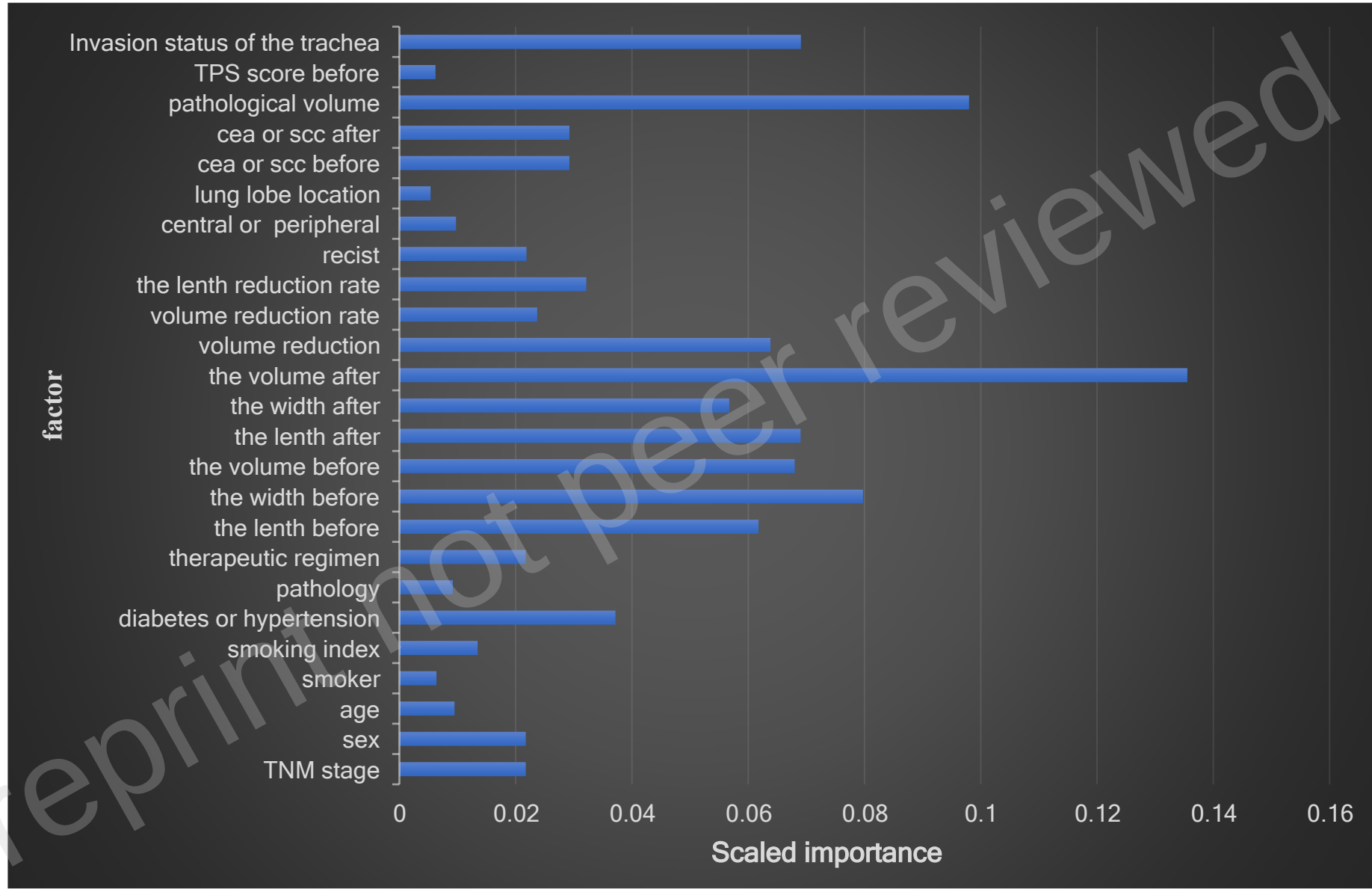


Figure 6b

models	Average Accuracy
Gaussian Naive Bayes	0.68
K-Neighbors Classifier	0.88
Linear Discriminant Analysis	0.72
Support Vector Machines	0.88
Decision Tree Classifier	0.56
Logistic Regression	0.68

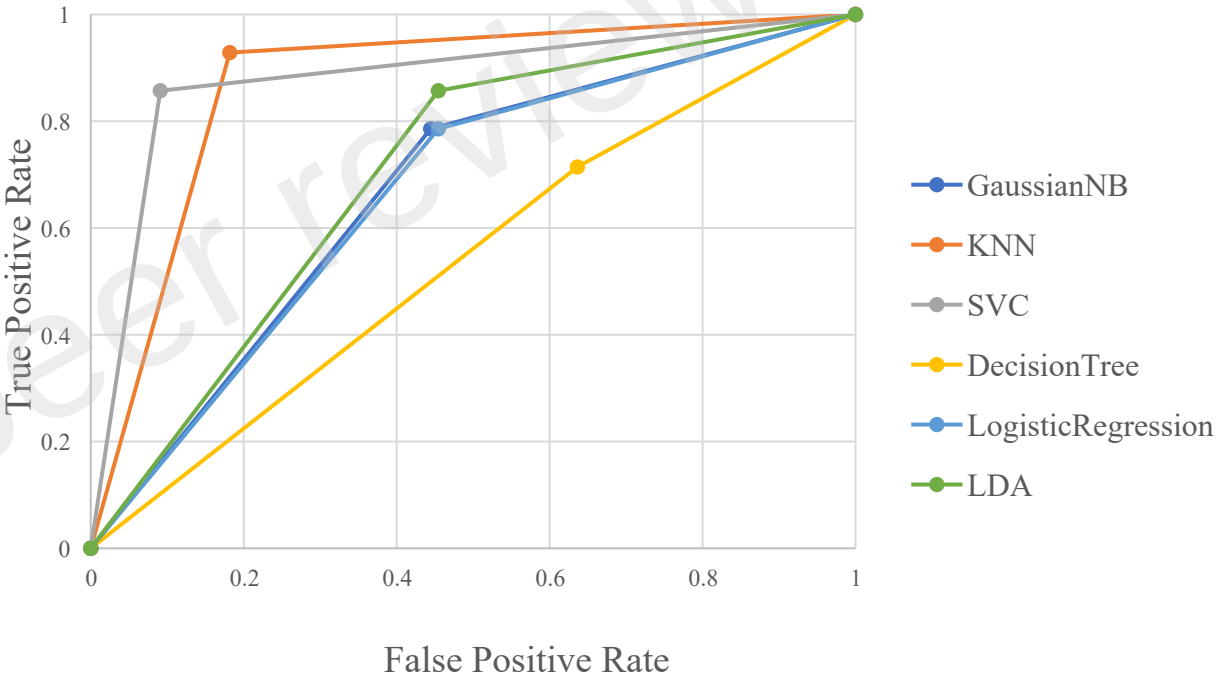


Figure 7a

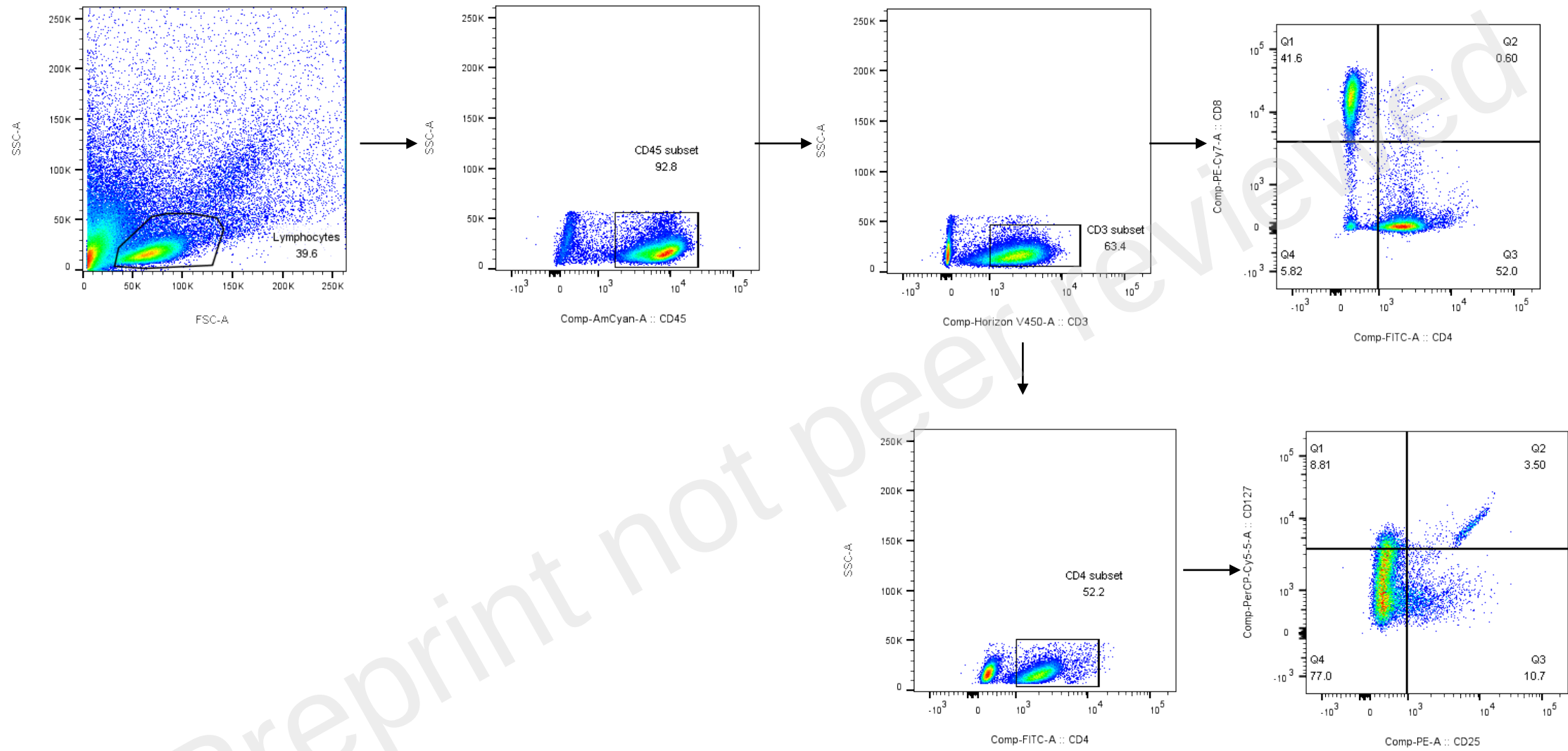


Figure 7b

